SCEPI: Single-Mutant-Conditioned Epistasis Prediction for Compensatory Protein Variant Discovery

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Abstract. Selecting compensatory double mutants requires predicting how individually damaging substitutions combine. We introduce Single-mutant-Conditioned Epistasis Prediction (SCEPI), a cross-protein model that conditions a shared interaction function on measured single-mutant effects. These measurements also calibrate individual effects outside the interaction network. Source combination measurements train the model, while target double-mutant labels remain withheld. Across 155 held-out proteins, SCEPI achieves mean residual root mean squared error of 0.595 kcal/mol, compared with 0.805 for ThermoMPNN-D, 0.847 for Kermut, and 0.853 for calibrated StabilityArc. At a budget of ten candidates per protein, its compensation hit rate is 11.22% across 120 proteins, compared with 7.07%, 5.40%, and 8.75%, respectively. This rate is 4.27 times the 2.63% expectation of uniform random selection from the same candidates. Matched input-removal controls retain external calibration and establish prediction and selection gains from measured single effects within the interaction network. The gains persist with the same measured single-effect sum, stricter homology grouping, and measurement-boundary analysis. SCEPI uses existing single-mutant measurements to guide compensatory discovery in unseen proteins.

1 Introduction

Compensatory substitutions recover stability or function lost through other sequence changes [1, 2, 5]. Their discovery requires selecting useful combinations from individually damaging substitutions. Measured single-mutant effects provide target-specific information for directing a fixed experimental budget toward those combinations.

Complete wild-type, single-mutant, and double-mutant measurements define observed nonadditivity [13]. Source proteins supply this supervision for learning a shared interaction function. At inference, target single-mutant measurements condition predictions for unmeasured combinations. Global phenotype nonlinearity and pair-dependent effects both contribute to the observed contrast [10, 11].

Transferable combination predictors learn from structure or sequence representations [3, 9, 14]. Target-specific regression also uses measured variants [6]. These capabilities motivate a question about the additional value of supplying target single-mutant measurements directly to a shared interaction function.

We introduce SCEPI, which conditions a shared interaction function on available single-mutant effects and their observation indicators. The same measurements calibrate individual effects outside that function. The pair function combines these inputs with reference and single-mutant representations. A double-mutant cycle cancels endpoint-only offsets and enforces zero interaction at either wild-type endpoint. Source combination measurements train the shared parameters.

Matched input-removal controls retain external calibration, model capacity, training samples, and selection procedures. Removing the measured pair inputs increases residual error and reduces compensation hit rate under both homology groupings. The full-target comparison gives SCEPI a top-10 hit rate of 11.22%, compared with 7.07% for ThermoMPNN-D, 5.40% for Kermut, and 8.75% for calibrated StabilityArc. Every

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method receives zero target double-mutant training labels. Fixing the measured single-effect sum and excluding measurements near assay limits preserve the selection gains. A same-capacity direct-head comparison tests the reference constraints.

2 Related work

Combination predictors from protein representations. ThermoMPNN-D uses a Siamese head over frozen ProteinMPNN features [3]. Mutate Everything composes reusable mutation representations [9]. ESM-MSR encodes wild-type and mutant sequences [12]. The StabilityArc preprint learns a shared wild-type decoder and a symmetric, contact-conditioned residual [14]. These models establish transferable combination prediction from protein representations.

Using target measurements. Table 1 distinguishes the roles of single-mutant measurements. Mutate Everything constructs training residuals from measured singles and uses predicted effects at inference [9]. StabilityArc supplies a cross-protein prior to Kermut's target Gaussian-process posterior [6, 14]. SCEPI supplies measured effects directly to its shared pair network and external calibration. The matched input-removal control isolates the pair-input role.

Phenotype maps and interaction terms. MAVE-NN, MoCHI, and reference-free analysis fit genetic effects and nonlinear phenotype maps [4, 11, 15]. Global epistasis separates phenotype nonlinearity from pair-dependent contributions [10]. SCEPI retains both in its predicted phenotype contrast. Component comparisons evaluate the map and fitting objectives separately from measured-input conditioning.

3 Methods

SCEPI combines a source-trained interaction function with measured target single effects (Figure 1). These measurements enter the pair network and external calibration. Target double-mutant outcomes are reserved for evaluation.

3.1 Prediction problem

Let s_p be a reference protein and M a set of substitutions at distinct sites. The predictor receives s_p, M, and the target measurements allowed by the information condition. It outputs the relative phenotype $\widehat{\mu}_p(M)$ and a candidate score. For two substitutions $A = (i, a)$ and $B = (j, b)$, the measured interaction is

$$\epsilon_p(A, B) = y_p(AB) - y_p(A) - y_p(B) + y_p(\varnothing). \tag{1}$$

The four observations form a double-mutant cycle [7]. They share a reference background, assay condition, and phenotype scale. Stability is expressed as unfolding free energy, with positive changes indicating increased stability. Target single-mutant labels are available in the main compensation task. Target test double-mutant labels evaluate predictions and selected candidates.

Table 1. Single-mutant information use. The final column records direct measured-value inputs to the pair network. ThermoMPNN-D target-head fitting refers to our single-only target protocol. Mutate Everything is included for the methodological comparison.

Method	Source use of single labels	Target-single use	Measured pair inputs
ThermoMPNN-D [3]	Data augmentation	Head adaptation	No
Mutate Everything [9]	Residual targets	Predicted effects	No
StabilityArc + Kermut [6, 14]	Decoder supervision	Target GP posterior	No
SCEPI	Joint loss	Calibration and conditioning	Effects and masks

3.2 Mutation representations and calibrated single effects

We use a frozen ESM-2 encoder h [8]. For substitution $A = (i, a)$, the endpoint representation contains $h_i(s_p)$, the local change $h_i(x_p(A)) - h_i(s_p)$, amino-acid identities, and the pretrained substitution score. A shared head predicts the single effect through a reference subtraction,

$$a_i(a) = f_\phi(u_i(a)) - f_\phi(u_i(w_i)), \tag{2}$$

where w_i is the reference residue. Thus $a_i(w_i) = 0$. The same head processes every protein.

Available single-mutant effects enter through

$$\widetilde{a}_i(a) = a_i(a) + \omega m_i(a)\big[\mu_i^{\mathrm{obs}}(a) - a_i(a)\big], \tag{3}$$

where $m_i(a)$ records label availability. The mixing weight ω is selected on source-validation proteins. A measured zero remains an available label. The calibrated effects enter both the single-effect sum and the interaction network.

Pair representations also contain mutation-dependent cross-site context and sequence separation. Appendix A defines these features and the structural variant.

3.3 Conditioning the shared interaction function

For a candidate pair, collect the calibrated single effects and their availability indicators in

$$c_{ij}(a, b) = \big(\widetilde{a}_i(a), \widetilde{a}_j(b), m_i(a), m_j(b)\big). \tag{4}$$

The shared network receives this vector together with the endpoint representations and site relation. Thus the pair function depends on the target's measured single effects while its parameters remain shared across proteins. Exchanging the endpoints also exchanges the entries of c_{ij}.

Let $\Psi_\theta(i, a; j, b \mid c_{ij})$ denote the ordered network. We average both orders:

$$F_\theta(i, a; j, b) = \tfrac{1}{2}\big[\Psi_\theta(i, a; j, b \mid c_{ij}) + \Psi_\theta(j, b; i, a \mid c_{ji})\big]. \tag{5}$$

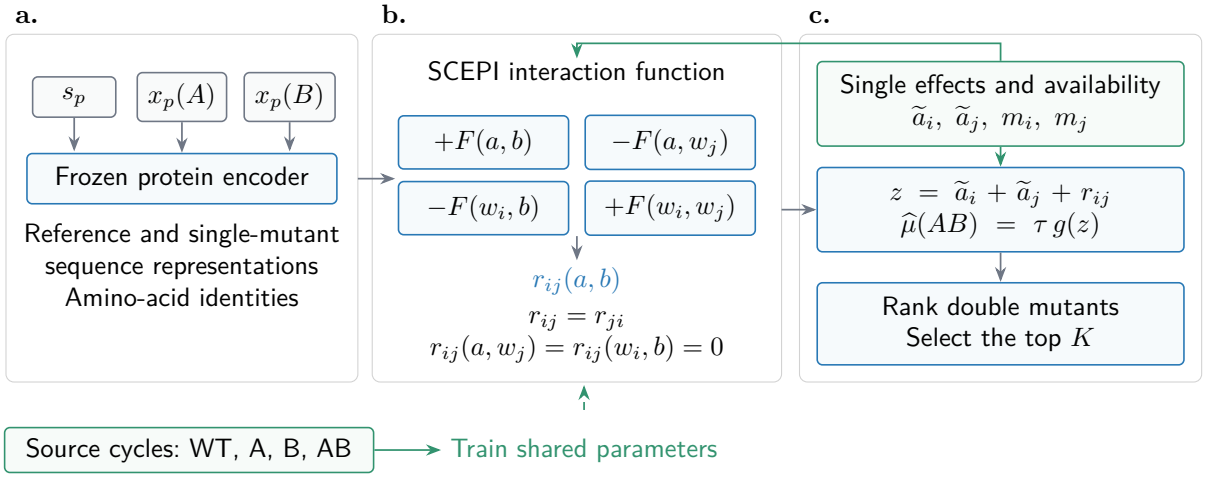


Fig. 1. SCEPI conditions shared interaction predictions on single-mutant measurements. (a) A frozen encoder supplies reference and mutation features from the reference sequence and its single mutants. (b) Calibrated single effects and their availability indicators condition every evaluation of the shared symmetric pair function. Subtraction around the double-mutant cycle enforces the reference constraints. (c) The same single effects enter the external sum before phenotype prediction. Double-mutant predictions then rank candidates at a fixed budget. Complete source cycles and source observations supervise SCEPI. Target double-mutant outcomes are reserved for evaluation.

The conditioning inputs enter every evaluation of F. The pair output uses the same subtraction over the four states,

$$r_{ij}(a,b) = F(i,a;j,b) - F(i,a;j,w_j) - F(i,w_i;j,b) + F(i,w_i;j,w_j). \tag{6}$$

Every call uses the specified amino-acid state. Restoring an endpoint to wild type sets its effect and local representation change to zero. It also removes the context change caused by that endpoint at the other site.

Equation (6) gives symmetry, $r_{ij}(a,b) = r_{ji}(b,a)$, and zero reference effects, $r_{ij}(a,w_j) = r_{ij}(w_i,b) = 0$. Replacing F by $F + v_i(a) + v_j(b) + c$ leaves r unchanged because the four terms cancel every endpoint-only contribution. A matched direct-head comparison tests the predictive contribution of these constraints.

3.4 Composition and phenotype prediction

For a mutation set M, the latent effect sums single and pair contributions:

$$z_p(M) = \sum_{(i,a)\in M} \widetilde{a}_i(a) + \sum_{\substack{(i,a),(j,b)\in M \\ i<j}} r_{ij}(a,b). \tag{7}$$

We predict $\widehat{\mu}_p(M) = \tau g(z_p(M))$, where $\tau > 0$, $g(0) = 0$, and $g'(0) = 1$. The monotone map models global phenotype nonlinearity. Its parameterization is given in the appendix.

The model's interaction on the observed phenotype scale is always calculated from its four predictions:

$$\widehat{\epsilon}_p(A,B) = \widehat{\mu}_p(AB) - \widehat{\mu}_p(A) - \widehat{\mu}_p(B). \tag{8}$$

This contrast includes global phenotype nonlinearity and the learned pair contribution. It equals r_{ij} under the identity map and unit scale. Appendix A gives the decomposition.

3.5 Learning from complete combination measurements

The single-effect head is initialized from source single-mutant measurements. Source combination measurements then jointly fit the single head, pair function, and phenotype map. Training combines a Huber loss on the complete predicted residual with the likelihood of source observations. Sampling balances proteins and site pairs. The likelihood retains point-valued and interval-valued measurements.

Source training varies the permitted single-mutant information to represent inference conditions. Source-validation proteins determine model configurations and the label-mixing weight. Target homology groups remain excluded from fitting and selection. Appendix C gives the full criterion, observation model, and implementation. Residual-only and phenotype-only variants compare objective choices.

3.6 Compensatory candidate selection

SCEPI ranks double mutants using its native phenotype prediction $\widehat{\mu}_p(AB)$. For a fixed budget K, it selects the highest-scoring K candidates. We evaluate recovery relative to the better constituent single mutant,

$$R_p(A,B) = y_p(AB) - \max\{y_p(A), y_p(B)\}. \tag{9}$$

A compensatory hit satisfies $R_p(A,B) \geq \delta$. The candidate task focuses on pairs whose two measured single effects are each at most $-\delta$.

To isolate interaction information, we fix the measured single-effect sum across methods. Each method m ranks the same candidates by

$$s_m(A,B) = \mu_p^{\mathrm{obs}}(A) + \mu_p^{\mathrm{obs}}(B) + \widehat{\epsilon}_{m,p}(A,B). \tag{10}$$

Every method retains its own four-prediction contrast. This comparison fixes the single-effect sum and changes the predicted nonadditivity. Native ranking measures the complete method's utility. Equation (10) tests the contribution of interaction predictions to that utility.

Uniform random selection samples K candidates without replacement from the same protein-level pool. Its expected hit rate is the pool's hit fraction, and its expected mean recovery is the pool mean. Enrichment is the ratio of the model and random equal-protein mean hit rates. Appendix K gives the exact calculation and uncertainty analysis.

4 Experimental setting

We use the cDNA-display proteolysis stability measurements of Tsuboyama et al. [13]. The accepted double-mutant set contains 182,663 variants across 190 reference backgrounds. It supplies 127,892 complete, point-valued double-mutant cycles across 160 backgrounds and 488 site pairs. The retrospective evaluation requires all four measurements to lie inside the assay range of $[-1, 5]$ kcal/mol. Model training also retains interval-valued observations through the observation likelihood.

The primary evaluation holds out homology groups and scores all accepted target double mutants. Grouping uses 30% sequence identity and at least 80% bidirectional coverage. Related reference backgrounds remain together, and annotated family relations further merge groups. Source training and validation exclude the outer test partition. S-all supplies all permitted target single-mutant labels and excludes every target double-mutant label from fitting. Homology groups and annotated families have separate coverage records.

Comparators include independently trained additive and global nonlinear models, ThermoMPNN-D, Kermut, StabilityArc + Kermut, mean ridge, and mean MLP. The global model retains the monotone map and omits the pair term. The mean-representation controls fit ridge regression or a multilayer perceptron (MLP) to averaged mutation representations. Source-trained baselines are retrained on the same source partitions, with checkpoints selected by source validation. StabilityArc scores supply the prior mean for Kermut target calibration [6, 14]. All methods receive zero target double-mutant labels and score the same candidates. Appendices C and D give their architectures, source supervision, target adaptation, and structural inputs.

Prediction error uses root mean squared error (RMSE). Phenotype evaluation requires at least 30 point-valued target measurements per protein. It covers 159 proteins and 51 homology groups. Residual evaluation requires at least 30 complete, in-range cycles and covers 155 proteins from the same 51 groups. Compensation selection requires at least 100 eligible candidates per protein and reference stability at most 4.5 kcal/mol. Its cohort contains 120 proteins and 44 groups. We use candidate budgets $K \in \{1, 3, 10, 30\}$ and recovery threshold $\delta = 0.5$ kcal/mol. Metrics are calculated within each protein, averaged over matched repetitions, and then averaged equally across proteins. Hyperparameters and label-mixing weights are chosen using source-validation measurements.

For each cycle, the margin is the smallest distance of any measurement from either assay limit. Margins of 0.25 and 0.5 kcal/mol retain 120 and 115 compensation proteins; the latter covers 43 groups. The designated site-pair subset and target combination adaptation have separate evaluations. Statistical and repetition details appear in the appendix.

Additional information conditions measure the value of target labels. Z permits no target measurements. S32 and S128 permit the reference and the specified number of target single-mutant labels. The C conditions additionally permit 128, 512, 2,048, or all available training double-mutant labels. Their designated site-pair holdout excludes test pairs from target combination fitting. Separate adaptation experiments cover six functional assays and nine FLIP2 split protocols [17]. Appendix I defines their measurement units, model variants, and evaluation populations.

5 Results

5.1 Measured single effects improve interaction prediction and selection

Conditioning the pair function on measured single effects improves prediction and compensation selection under both homology groupings (Table 2; Figure 4c,d). The matched input-removal control supplies model-predicted single effects and zero availability indicators to F. It retains the external calibration formula and source-validation selection procedure. Model capacity, initialization, training samples, objectives, and stopping rules are matched. Each model is refitted independently.

Measured pair inputs reduce residual RMSE by 22.4% under the primary grouping and 25.7% under the stricter grouping. Native top-10 hit rates improve by 3.03 and 3.00 percentage points. Both mean gains are positive in every matched training repetition under each grouping. The residual and compensation cohorts contain the same 155 and 120 proteins across controls.

Table 2. Matched pair-function comparisons under 30% and 25% homology grouping. Residual prediction covers 155 proteins, and native top-10 selection covers 120. The predicted-singles control removes measured values and availability indicators from F while retaining external calibration. The direct head keeps those inputs and replaces cycle subtraction with the same network's direct output.

	Residual RMSE (kcal/mol)		Native hit rate (%)	
Pair function	30%	25%	30%	25%
SCEPI	**0.595**	**0.604**	**11.22**	11.15
Predicted singles	0.767	0.814	8.18	8.15
Direct residual head	0.635	0.630	10.70	**11.38**

Table 3. Full-target prediction and native top-10 compensation selection in held-out homology groups. Phenotype, residual, and selection cohorts contain 159, 155, and 120 proteins. RMSE uses kcal/mol. Values are equal-protein means over matched repetitions. Enrichment is relative to uniform random selection from the same candidates. All models receive zero target double-mutant training labels.

Predictor	Phenotype RMSE	Residual RMSE	Hit rate (%)	Recovery (kcal/mol)	Enrichment (×)
SCEPI	**0.657**	**0.595**	11.22	−0.257	4.27
ThermoMPNN-D	0.777	0.805	7.07	−0.440	2.69
Kermut	0.798	0.847	5.40	−0.458	2.05
StabilityArc + Kermut	0.815	0.853	8.75	−0.344	3.33
Global nonlinear	0.911	0.635	4.45	−0.437	1.69
Additive	0.893	1.091	6.38	−0.387	2.43
Mean MLP	1.405	0.966	10.07	−0.413	3.83
Mean ridge	1.328	1.002	**11.42**	−0.354	**4.34**
Uniform random	—	—	2.63	−0.900	1.00

The input contribution remains when every method uses the same measured single-effect sum. Hit-rate gains over the input-removal control are 3.08 and 3.02 percentage points under the two groupings. At the 0.5 kcal/mol measurement margin, the gains are 3.11 and 3.13 points. These comparisons identify selection information supplied by measured inputs within the interaction function.

5.2 The conditional predictor improves compensatory candidate discovery

SCEPI achieves a mean top-10 compensation hit rate of 11.22% using its native predictions (Table 3). ThermoMPNN-D achieves 7.07%, Kermut 5.40%, and the global nonlinear model 4.45%. SCEPI improves their hit rates by 4.15, 5.82, and 6.77 percentage points. Its mean selected recovery exceeds these methods by 0.183, 0.201, and 0.180 kcal/mol. Recovery is higher in 86 of 120 proteins against ThermoMPNN-D and in 93 of 120 against Kermut. Figure 2 shows the full budget curves and all protein-level hit-rate differences.

SCEPI has the highest mean selected recovery among the seven baselines at every evaluated budget. At $K = 10$, its recovery advantage over mean ridge is 0.098 kcal/mol. Mean ridge has a hit rate of 11.42%, compared with SCEPI's 11.22% and mean MLP's 10.07%. Against the additive model, SCEPI improves hit rate by 4.83 percentage points and mean recovery by 0.130 kcal/mol. The additive hit rate is 6.38%.

SCEPI improves top-10 hit rate over StabilityArc + Kermut by 2.47 percentage points and mean recovery by 0.087 kcal/mol. StabilityArc + Kermut reaches an 8.75% hit rate. SCEPI gives higher recovery in 73 of 120 proteins. Both methods learn pair effects across source proteins and use target single-mutant measurements.

SCEPI enriches compensatory hits 4.27-fold over uniform random selection at $K = 10$. The same 120 proteins contain 79,564 eligible candidates. Random selection has an expected equal-protein hit rate of 2.63% and mean recovery of −0.900 kcal/mol. SCEPI increases the hit rate by 8.59 percentage points and mean recovery by 0.643 kcal/mol. Enrichment is 5.52, 4.90, 4.27, and 3.26 at budgets 1, 3, 10, and 30, respectively (Figure 2a,b).

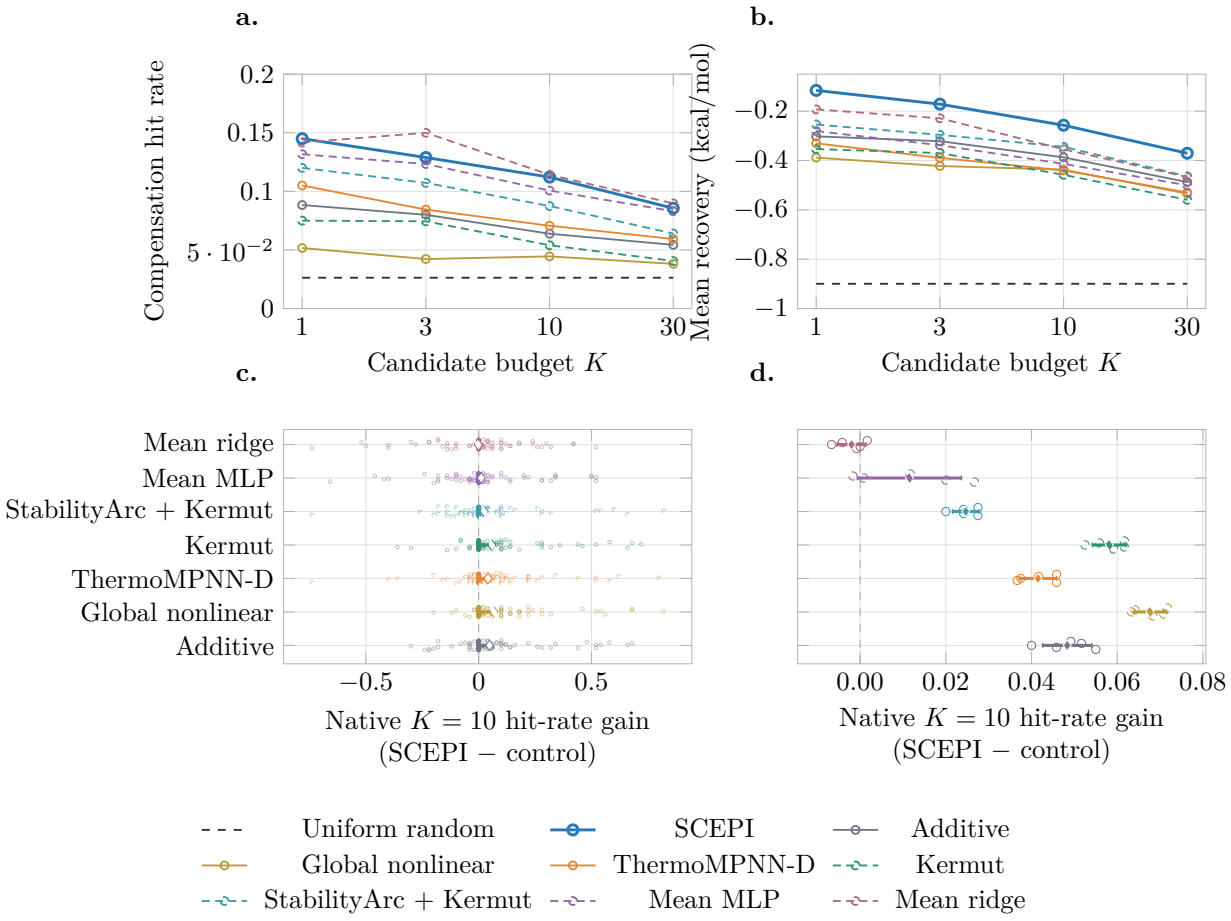


Fig. 2. SCEPI compensatory discovery in full-target S-all evaluation. (a) Native hit rate and (b) mean recovery across candidate budgets. Dashed black lines show exact uniform-random expectations from the same candidates. (c) Protein-level top-10 hit-rate gains after averaging repetitions within each of 120 proteins. Open points denote proteins; segments, ticks, and diamonds denote interquartile ranges, medians, and means. (d) Repetition-level gains in the same control order. Open points are equal-protein means; diamonds and whiskers show their mean and standard deviation. All predictors use identical candidates and native phenotype scores. StabilityArc + Kermut uses source-held-out scores with target calibration.

5.3 The learned function predicts interactions in unseen combinations

Mean residual RMSE is 0.595 kcal/mol for SCEPI, 0.805 for ThermoMPNN-D, and 0.847 for Kermut. The respective error reductions are 26.1% and 29.8%. SCEPI has lower protein-level residual RMSE in 136 of 155 proteins against either method (Figure 3a).

SCEPI reduces residual RMSE by 30.3% relative to StabilityArc + Kermut, which reaches 0.853 kcal/mol. Its error is lower in 130 of 155 proteins.

SCEPI reduces the global nonlinear control's mean residual error by 6.4%, with lower error in 92 of 155 proteins. The global control reaches 0.635 kcal/mol, compared with 1.091 kcal/mol for the additive model. This comparison separates phenotype nonlinearity from the additional pair function.

SCEPI's phenotype RMSE is 0.657 kcal/mol. Its mean Spearman correlation is 0.742. ThermoMPNN-D reaches 0.777 and 0.673, while Kermut reaches 0.798 and 0.696. StabilityArc + Kermut reaches 0.815 and 0.746. SCEPI has lower phenotype RMSE in 117 of 159 proteins against ThermoMPNN-D and in 114 of 159 against Kermut. Its mean phenotype and residual errors also improve over both target-fitted mean-representation baselines. Figure 3b–d shows the protein-level differences and repetition-level residual gains.

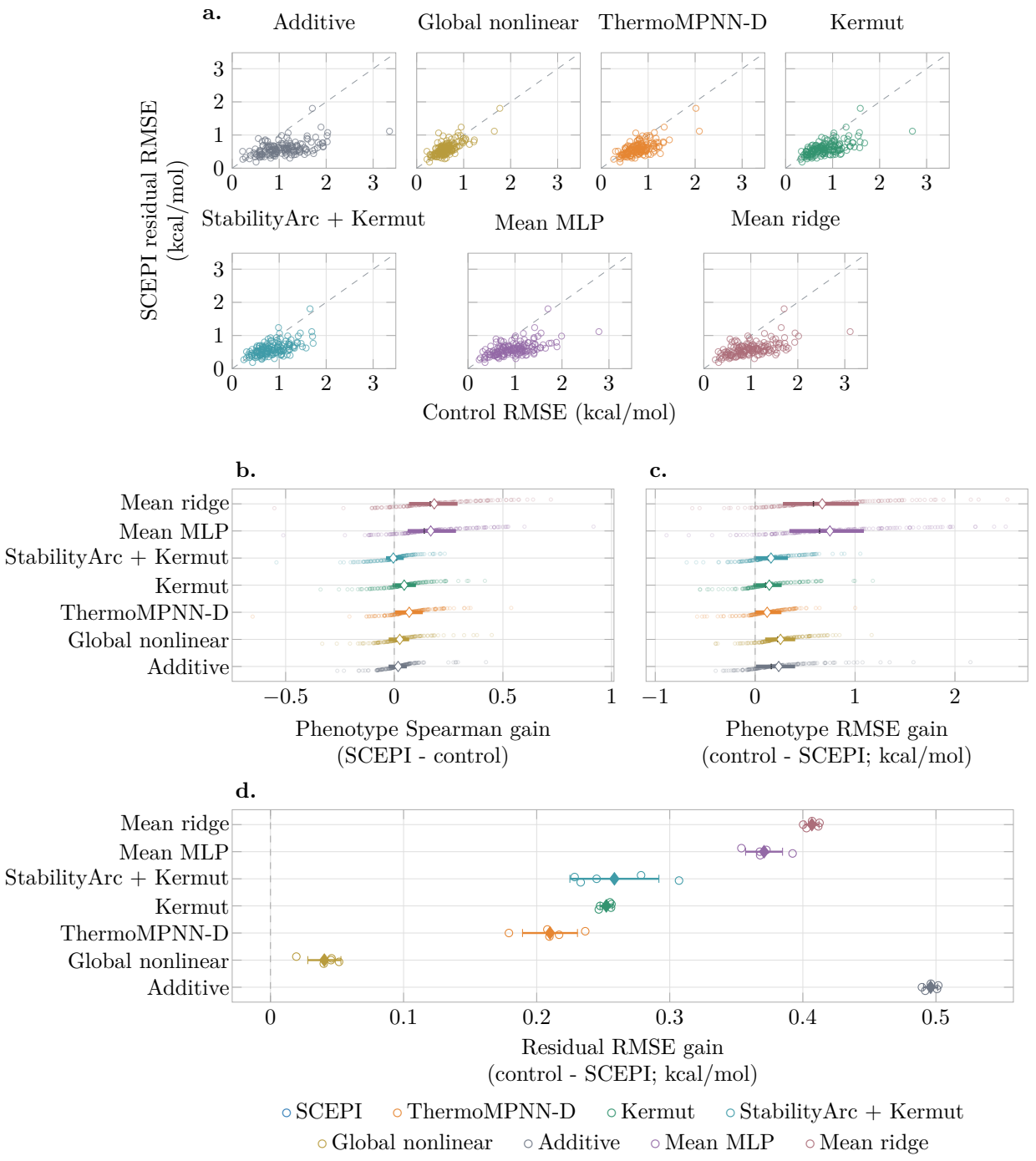


Fig. 3. SCEPI prediction across held-out homology groups in the full-target S-all setting. (a) Paired residual RMSE for all 155 eligible proteins. The dashed diagonal denotes equal error. (b) Native phenotype Spearman gain across 159 proteins. (c) Phenotype RMSE gain in the same cohort. In (b) and (c), points are proteins, segments are interquartile ranges, dark ticks are medians, and diamonds are means. (d) Repetition-level residual RMSE gains. Open points are equal-protein macro averages; diamonds and whiskers show the mean and standard deviation. Positive gains favor SCEPI. All eligible proteins are retained.

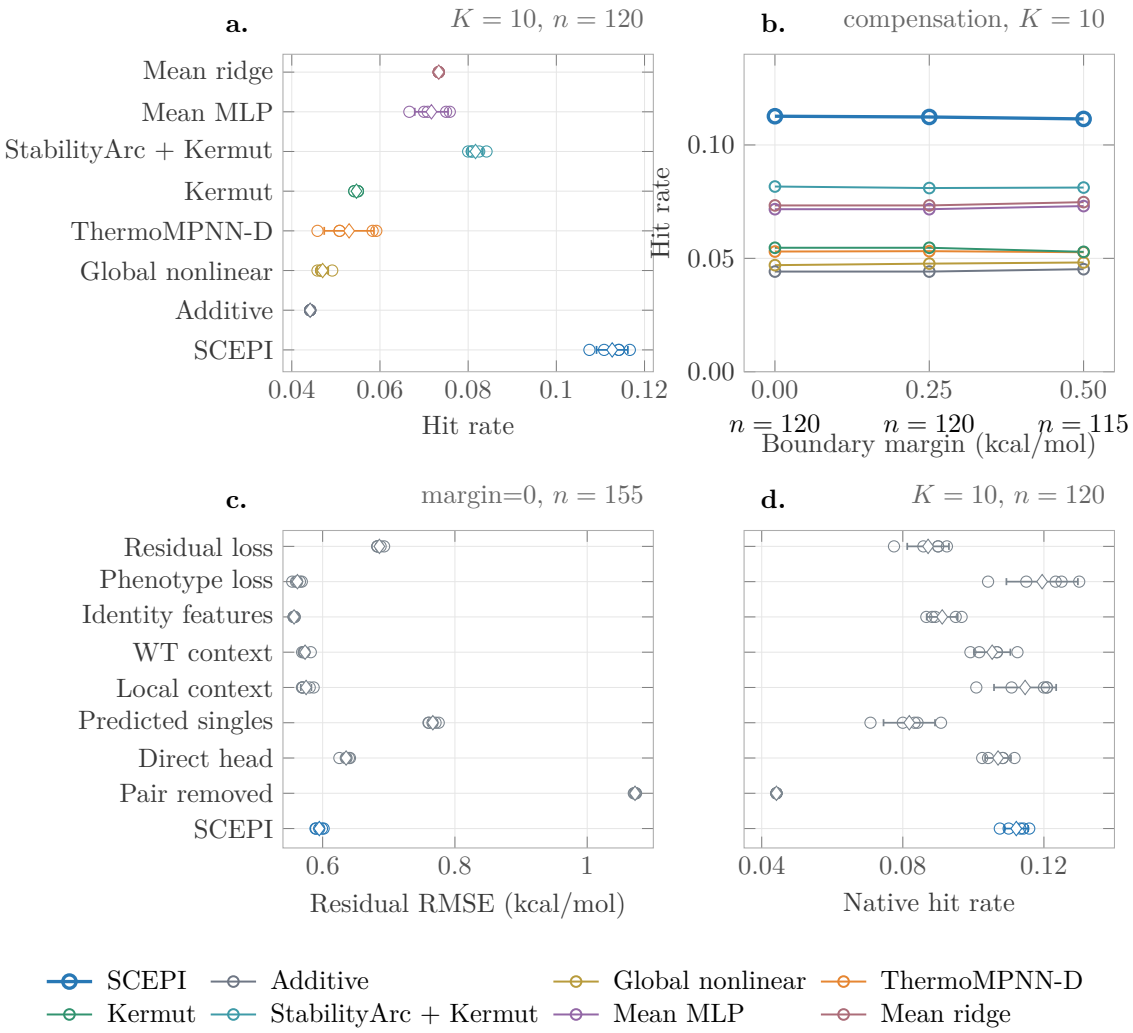


Fig. 4. Measured-single conditioning and component comparisons under 30% identity grouping. (a) Top-10 hit rates with the same measured single-effect sum. (b) Sensitivity to distance from assay limits; cohort sizes appear below the axis. (c) Residual RMSE and (d) native hit rate share the labeled control order. The direct head has matched capacity. The input-removal control retains external calibration and removes measured effects and availability from F. Open points are equal-protein repetition means; diamonds and whiskers show their mean and standard deviation. Residual and selection cohorts contain 155 and 120 held-out proteins.

With 128 target double-mutant training labels, the source-initialized SCEPI adaptation variant reaches unseen-site-pair RMSE of 0.567 kcal/mol across 70 proteins. Kermut reaches 0.758, ThermoMPNN-D 0.775, second-order MoCHI 0.859, and the target-only SCEPI adaptation variant 0.799. Table 13 reports every comparator across the four budgets.

5.4 Interaction predictions add value beyond measured single effects

Fixing the measured single-effect sum across methods preserves the selection advantage (Figure 4a; Appendix Table 7). SCEPI achieves a compensation hit rate of 11.27%, compared with 4.42% for additivity and 4.70% for global nonlinearity. ThermoMPNN-D reaches 5.30%, Kermut reaches 5.47%, and StabilityArc + Kermut reaches 8.17%. Mean MLP and mean ridge reach 7.17% and 7.33%. Mean selected recovery improves by 0.176, 0.161, 0.224, and 0.200 kcal/mol over the first four controls. Its improvement over StabilityArc + Kermut is 0.095 kcal/mol. These paired scores differ through each method's predicted four-term nonadditivity.

At a margin of at least 0.5 kcal/mol, SCEPI's hit rate is 11.15%. Additivity reaches 4.52%, global nonlinearity 4.82%, ThermoMPNN-D 5.27%, and Kermut 5.29%. StabilityArc + Kermut reaches 8.12%. Mean MLP and mean ridge reach 7.30% and 7.48%. The advantage extends to combinations whose measurements are separated from both assay limits (Figure 4a,b).

5.5 Reference constraints improve residual prediction

Cycle subtraction improves mean residual accuracy against the same-capacity direct head. The direct head's residual RMSE is 0.635 kcal/mol under the primary grouping and 0.630 kcal/mol under the stricter grouping. SCEPI's corresponding errors are 0.595 and 0.604 kcal/mol. Its native hit-rate gain is 0.52 percentage points under the primary grouping and -0.23 points under the stricter grouping. This parameterization supplies exact reference constraints and a residual-prediction benefit across both groupings. Tables 8 and 9 report the prediction and selection results.

Removing the learned pair term at inference retains the trained single head and phenotype map. Native hit rate falls from 11.22% to 4.42%, and residual RMSE rises from 0.595 to 1.072 kcal/mol. This intervention establishes the pair output's contribution within the trained predictor (Figure 4c,d). Representation, mapping, and training-objective variants are reported with their complete results in the appendix.

6 Discussion

SCEPI makes existing single-mutant measurements useful for selecting compensatory combinations in unseen proteins. Source combination measurements train a shared function, and target single effects condition its predictions. This information supports candidate selection before target combination labels are collected. The retrospective comparisons quantify that selection benefit at a fixed budget and across held-out homology groups.

Measured inputs extend the use of target labels beyond fitting single effects or a Gaussian-process posterior. StabilityArc supplies a source-trained prior for target adaptation [14]. SCEPI also uses the observed single effects within the shared interaction function. The matched removal experiment identifies the value of that input location while retaining external calibration. Selection gains persist when every method uses the same measured single-effect sum.

The reference constraints and the measured inputs serve distinct roles. The constraints define zero interaction at a wild-type endpoint and improve mean residual accuracy against a direct head. Their hit-rate differences vary across groupings. Conditioning on measured target observations improves both prediction and compensatory selection under the two homology groupings.

7 Conclusion

SCEPI conditions a transferable interaction function on measured single effects. Matched controls and held-out comparisons establish gains in residual prediction and compensatory discovery. Code, configurations, and reproduction materials are available at `https://github.com/SZU-ADDG/SCEPI`.

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